

21-3e Deriving the Demand Curve

We have just seen how changes in the price of a good alter the consumer's budget constraint and, therefore, the quantities of the two goods that she chooses to buy. The demand curve for any good reflects these consumption decisions because it shows the quantity demanded of a good for any given price. A consumer's demand curve is a summary of the optimal decisions that arise from her budget constraint and indifference curves.

For example, [Figure 12](#) considers the demand for Pepsi. Panel (a) shows that when the price of a liter falls from \$2 to \$1, the consumer's budget constraint shifts outward. Because of both income and substitution effects, the consumer increases her purchases of Pepsi from 250 to 750 liters. Panel (b) shows the demand curve that results from this consumer's decisions. In this way, the theory of consumer choice provides the theoretical foundation for the consumer's demand curve.

Figure 12

Deriving the Demand Curve

Panel (a) shows that when the price of Pepsi falls from \$2 to \$1, the consumer's optimum moves from point A to point B, and the quantity of Pepsi consumed rises from 250 to 750 liters. The demand curve in panel (b) reflects this relationship between the price and the quantity demanded.

It may be comforting to know that the demand curve arises naturally from the theory of consumer choice, but this exercise by itself does not justify developing the theory. There is no need for a rigorous, analytic framework just to establish that people respond to changes in prices. The theory of consumer choice is, however, useful in studying various decisions that people make as they go about their lives, as we see in the next section.

QuickQuiz

5. Bart and Lisa are both optimizing consumers in the markets for shirts and hats, where they pay **\$100** for a shirt and **\$50** for a hat. Bart buys 8 shirts and 4 hats, while Lisa buys 6 shirts and 12 hats. From this information, we can infer that Bart's marginal rate of substitution is ____ hats per shirt, while Lisa's is ____ .
- a. 2; 1
 - b. 2; 2
 - c. 4; 1
 - d. 4; 2
6. Maggie buys peanut butter and jelly, both of which are normal goods. When the price of peanut butter rises, the income effect induces Maggie to buy ____ peanut butter and ____ jelly.
- a. more; more
 - b. more; less
 - c. less; more
 - d. less; less
7. Ned buys wine and bread. When the price of wine rises, the substitution effect induces Ned to buy ____ wine and ____ bread.
- a. more; more
 - b. more; less
 - c. less; more
 - d. less; less

Printed By: Mike Benton (mbento6@wgu.edu)
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